

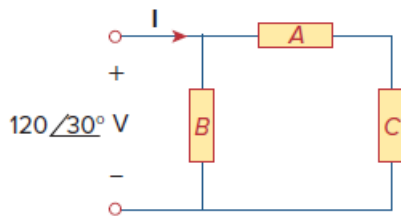
Problem

In the circuit of Fig. 11.72, load A receives 4 kVA at 0.8 pf leading. Load B receives 2.4 kVA at 0.6 pf lagging. Box C is an inductive load that consumes 1 kW and receives 500 VAR.

(a) Determine I.

(b) Calculate the power factor of the combination.

Figure 11.72:



Step-by-step solution

Step 1 of 8

Refer to Figure 11.72 in the textbook.

(a)

Calculate the real power of Box A.

$$P_A = S_A (pf_A)$$

Substitute 4 kVA for S_A and 0.8 for pf_A .

$$\begin{aligned} P_A &= (4 \times 10^3)(0.8) \\ &= 3200 \text{ W} \\ &= 3.2 \text{ kW} \end{aligned}$$

Step 2 of 8

Calculate the reactive power of Box A.

$$Q_A = \sqrt{S_A^2 - P_A^2}$$

Substitute 3.2 kW for P_A and 4 kVA for S_A .

$$\begin{aligned} Q_A &= \sqrt{(4000)^2 - (3200)^2} \\ &= \sqrt{5760000} \\ &= 2400 \text{ VAR} \\ &= 2.4 \text{ kVAR} \end{aligned}$$

Calculate the complex power of Box A.

$$S_A = P_A - jQ_A \quad (-ve \text{ indicates leading})$$

Substitute 3.2 kW for P_A and 2.4 kVAR for Q_A .

$$S_A = 3200 - j2400 \text{ VA}$$

Step 3 of 8

Calculate the real power of Box B.

$$P_B = S_B (pf_B)$$

Substitute 2.4 kVA for S_B and 0.6 for pf_B .

$$\begin{aligned} P_B &= (2.4 \times 10^3)(0.6) \\ &= 1440 \text{ W} \\ &= 1.44 \text{ kW} \end{aligned}$$

Step 4 of 8

Calculate the reactive power of Box B.

$$Q_B = \sqrt{S_B^2 - P_B^2}$$

Substitute 1.44 kW for P_B and 2.4 kVA for S_B .

$$\begin{aligned} Q_B &= \sqrt{(2400)^2 - (1440)^2} \\ &= \sqrt{3686400} \\ &= 1920 \text{ VAR} \\ &= 1.92 \text{ kVAR} \end{aligned}$$

Calculate the complex power of Box B.

$$S_B = P_B + jQ_B \quad (+ \text{ve indicates lagging})$$

Substitute 1.44 kW for P_B and 1.92 kVAR for Q_B .

$$S_B = 1440 + j1920 \text{ VA}$$

Step 5 of 8

Calculate the complex power of Box C.

$$S_C = P_C + jQ_C$$

Substitute 1000 W for P_C and 500 VAR for Q_C .

$$S_C = 1000 + j500 \text{ VA}$$

Calculate the total complex of the circuit.

$$S_t = S_A + S_B + S_C$$

Substitute $3200 - j2400 \text{ VA}$ for S_A , $1440 + j1920 \text{ VA}$ for S_B and $1000 + j500 \text{ VA}$ for S_C .

$$\begin{aligned} S_t &= 3200 - j2400 + 1440 + j1920 + 1000 + j500 \\ &= 5640 + j20 \\ &= 5640.035 \angle 0.203^\circ \text{ VA} \end{aligned}$$

Step 6 of 8

Calculate the current through the circuit, $\mathbf{I}$.

$$S_t = V_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

$$\mathbf{I}_{\text{rms}}^* = \frac{S_t}{V_{\text{rms}}}$$

Substitute $120\angle 30^\circ \text{ V}$ for V_{rms} and $5640.035\angle 0.203^\circ \text{ VA}$ for S_t .

$$\mathbf{I}_{\text{rms}}^* = \frac{5640.035\angle 0.203^\circ \text{ VA}}{120\angle 30^\circ \text{ V}}$$

$$= 47\angle -29.8^\circ \text{ A}$$

$$\mathbf{I}_{\text{rms}} = 47\angle 29.8^\circ \text{ A}$$

Thus, the current through the circuit is $\boxed{47\angle 29.8^\circ \text{ A}}$.

Step 7 of 8

(b)

Consider the value of total complex power of the circuit.

$$S_t = 5640.035\angle 0.203^\circ \text{ VA}$$

The phase angle is,

$$\theta = 0.203^\circ$$

Step 8 of 8

Calculate the power factor of the combination.

$$pf = \cos \theta$$

Substitute 0.203° for θ .

$$pf = \cos(0.203^\circ)$$

$$= 0.9999$$

$$\approx 1 \text{ (lagging)}$$

Thus, the power factor of the combination is $\boxed{1 \text{ (lagging)}}$.